# Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics | OpenStax: OpenStax Viewer

COURSE AT A GLANCE

### Review Content

REVIEW

**Primary:** Intermediate Algebra 2e 3.2 Slope of a Line; Intermediate Algebra 2e 3.1 Graph Linear Equations in Two Variables; College Algebra 2e 4.1 Linear Functions; Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables; Intermediate Algebra 2e 5.2 Properties of Exponents and Scientific Notation

**UNIT LEARNING OBJECTIVES**

I can determine and interpret the slope of a line.

I can graph a line from an equation and write an equation from given information.

I can apply a linear function to a context.

I can identify the solution of a linear system as an intersection point.

I can solve a simple two-variable linear system using an appropriate method.

I can simplify an expression using basic positive-integer exponent rules.

### Functions and Function Interpretation

REQUIRED

**Primary:** Intermediate Algebra 2e 3.5 Relations and Functions; College Algebra 2e 3.3 Rates of Change and Behavior of Graphs **Supplemental:** Intermediate Algebra 2e 3.6 Graphs of Functions; College Algebra 2e 3.1 Functions and Function Notation; College Algebra 2e 4.1 Linear Functions

**UNIT LEARNING OBJECTIVES**

I can evaluate a function from a formula, graph, or table.

I can determine whether a relation is a function.

I can determine domain and range from a graph or table.

I can solve for an input that produces a specified output.

I can interpret inputs, outputs, domain, and range in context.

I can calculate and interpret average rate of change.

I can compare functions represented by formulas, graphs, or tables.

### Polynomials and Factoring

REQUIRED

**Primary:** 5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products; 6.5 Polynomial Equations

**UNIT LEARNING OBJECTIVES**

I can add and subtract polynomials.

I can multiply polynomials.

I can factor out a greatest common factor.

I can factor a quadratic trinomial.

I can factor a difference of squares and other required special cases.

I can use factoring to determine the zeros of a polynomial.

### Rational Expressions and Equations

REQUIRED

**Primary:** 7.1 Multiply and Divide Rational Expressions; 7.2 Add and Subtract Rational Expressions; 7.4 Solve Rational Equations

**UNIT LEARNING OBJECTIVES**

I can state excluded values for a rational expression.

I can simplify a rational expression while preserving its restrictions.

I can multiply and divide rational expressions.

I can add and subtract rational expressions using a common denominator.

I can solve a rational equation and reject extraneous solutions.

## Exponents and Radicals

REQUIRED

**Primary:** Intermediate Algebra 2e 8.3 Simplify Rational Exponents; Intermediate Algebra 2e 8.2 Simplify Radical Expressions; Intermediate Algebra 2e 8.6 Solve Radical Equations
**Supplemental:** College Algebra 2e 2.6 Other Types of Equations

### UNIT LEARNING OBJECTIVES

I can simplify expressions containing negative exponents.

I can rewrite expressions between radical and rational-exponent forms.

I can simplify radical expressions.

I can apply exponent rules to expressions with integer and rational exponents.

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

## Quadratic Functions and Equations

REQUIRED

**Primary:** 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 6.5 Polynomial Equations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.5 Solve Applications of Quadratic Equations
**Supplemental:** 9.6 Graph Quadratic Functions Using Properties; 6.5 Polynomial Equations

### UNIT LEARNING OBJECTIVES

I can identify key features of a quadratic function from its graph or equation.

I can connect standard, factored, and vertex forms to features of a parabola.

I can solve a quadratic equation by factoring.

I can solve a quadratic equation using the square-root property.

I can solve a quadratic equation by completing the square.

I can solve a quadratic equation using the quadratic formula.

I can choose an efficient method for solving a quadratic equation.

I can graph a quadratic function using its vertex, intercepts, and symmetry.

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

## Extension Content

EXTENSION

**Primary:** 8.8 Use the Complex Number System; 9.3 Solve Quadratic Equations Using the Quadratic Formula

### UNIT LEARNING OBJECTIVES

I can perform basic arithmetic with complex numbers.

I can express the nonreal solutions of a quadratic equation using complex numbers.